



Pharm*Achieve*

ACUTE SINUSITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

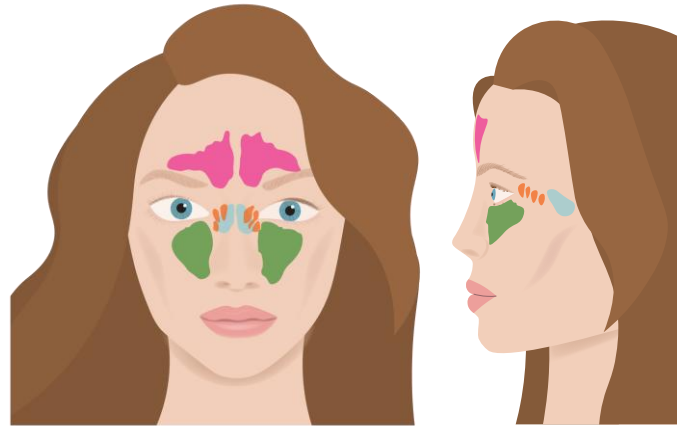
Professionalism

LEGEND

Abx	Antibiotic
FQ	Fluoroquinolones
INCS	Intranasal Corticosteroids
PO	By Mouth
SMX/TMP	Sulfamethoxazole/Trimethoprim
TID	Three times daily

Licensed to Shreyansh Salunke (ID: 4477)

WHAT IS SINUSITIS?



- - Frontal sinus
- - Sphenoid sinus
- - Ethmoid sinus
- - Maxillary sinus

- Inflammation of the paranasal sinuses and is a subcategory of upper respiratory tract infections
- Causes may be allergic, viral, bacterial, or fungal

Acute sinusitis

- Lasting ≤ 4 weeks
- Viral >>> bacterial (*S. pneumoniae*, *H. influenzae*, *M. catarrhalis*)

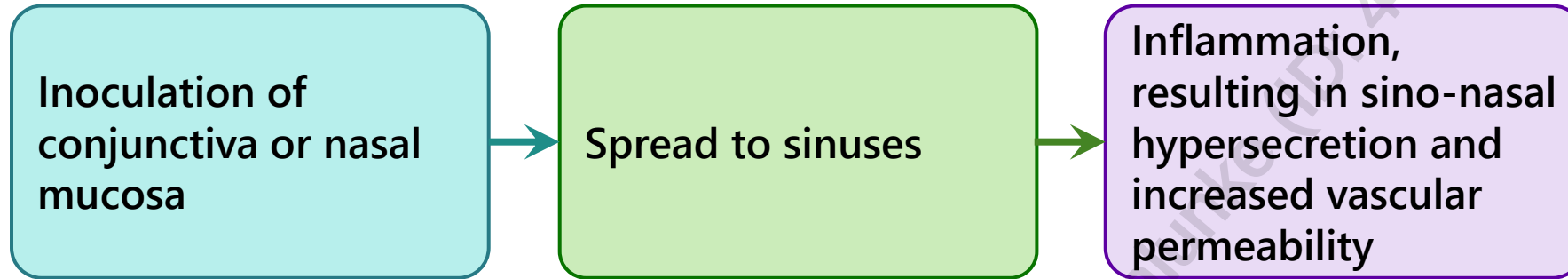
Chronic sinusitis infection

- Lasting ≥ 12 weeks
- *S. aureus*, enteric gram-negative bacteria, *Pseudomonas spp*, fungal

Recurrent sinusitis

- ≥ 4 or more episodes of acute sinusitis per year
- Each episode lasting at least 10 days

PATHOPHYSIOLOGY AND RISK FACTORS



- Usually **viral** origin
- Rarely complicated by **bacterial** infections
- Symptoms are similar between bacterial and viral sinusitis infections
 - Differentiation of **viral** and bacterial is based on type and duration of symptoms
 - **Viral infections peak at 3 days, improve in 7-10 days**
 - **Bacterial infections last > 10 -12 days and/or worsen after 3-4 days**
- **Risk factors: preceding viral respiratory tract infection, prior recurrent rhinosinusitis,**
- **risk factors for penicillin-resistant *S. pneumoniae* (daycare attendance, recent antibiotic use, age <2 years, age >65 years, immunocompromised)**

CLINICAL PRESENTATION OF ACUTE BACTERIAL SINUSITIS

Clinical presentation in adults	Clinical presentation in children	Note, chronic sinusitis presents with less specific symptoms than acute sinusitis
• Pain, pressure or fullness in the sinuses • Obstruction (nasal) • Discharge, purulent nasal/postnasal • Smell, hyposmia/anosmia • Other • Headache (mild) • Dental pain • Fatigue • Fever • Ear pain/pressure	• Nasal discharge • Fever • Facial or sinus swelling and/or pain • Irritability • Lethargy • Cough • Vomiting • When symptoms last >7-10 days or symptoms are severe (fever > 39°C, severe pain), antibiotics are indicated	Most symptoms resolve without medical therapy, however, if symptoms worsen and are left untreated, rare complications may include chronic sinusitis, spread of infection (resulting in meningitis, osteomyelitis, sepsis)

Complicated clinical presentation: changes in mental status, immunocompromised, significant comorbidities, risk factors for beta-lactam resistance/antibiotic failure, isolated frontal sinusitis, intense periorbital swelling/redness, and facial pain

DIAGNOSIS

Diagnosis

- Based on clinical findings from patient history and physical examination

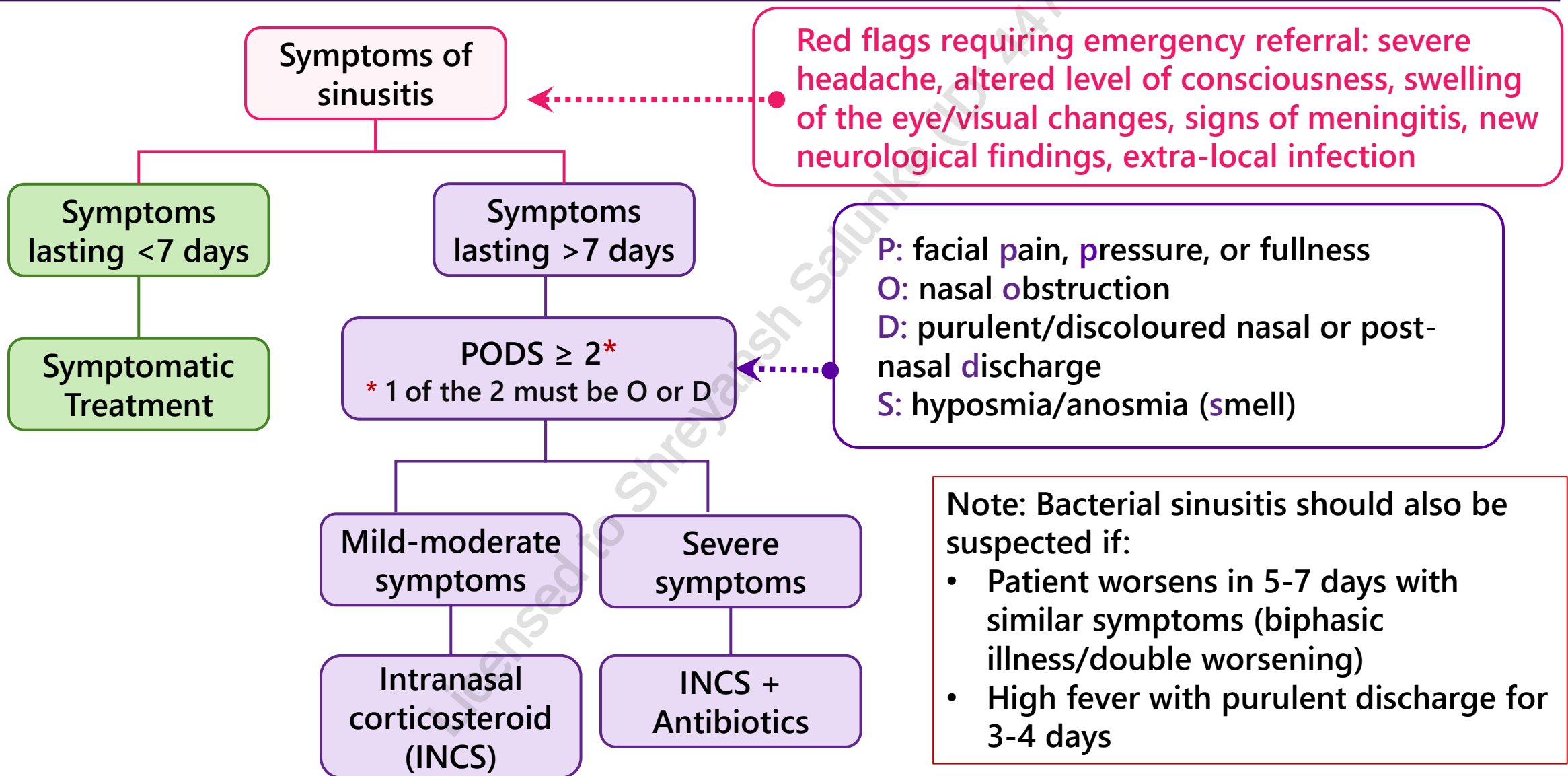
Differential diagnosis

- Allergic rhinitis, asthma, bronchitis
- Headaches (cluster, tension, migraine)
- Dental pain, otitis media

GOALS OF THERAPY

- ✓ Comforting patient and supportive care
- ✓ Encouraging hygiene for prevention of disease spread
- ✓ Providing symptomatic relief for congestion, pain, and fever
- ✓ Providing antibiotic treatment for bacterial infections

TREATMENT ALGORITHM



NON-PHARMACOLOGICAL MEASURES

Irrigation with nasal saline for congestion

Rest and hydration

Regular handwashing to limit infection spread

Avoid allergen exposure and environmental toxins
e.g., tobacco smoke, pollen

Sleep with head of bed elevated

Apply warm facial compresses

PHARMACOLOGICAL MEASURES FOR SYMPTOMATIC RELIEF

Decongestants

- Intranasal oxymetazoline or xylometazoline for nasal congestion (Limit use to 3 days)
- Oral decongestants may be useful for Eustachian tube dysfunction (proceed with caution for patients with cardiovascular disease, hypertension, glaucoma, etc.)
- Avoid in children <6 years of age due to side effect profile

Antihistamines

- anticholinergic side effects may dry the mucosa and limit clearance of secretions
- Second-generation antihistamines may be useful in allergic sinusitis

Nasal corticosteroids

- Some evidence for benefit if mild-moderate symptoms and as adjunct to antibiotics if severe symptoms
- Patients with chronic or allergic sinusitis may benefit more

Analgesics and antipyretics

- Can be used for pain and fever

ANTIBIOTIC THERAPY FOR SINUSITIS

Condition	1 st line therapy	2 nd line treatment
Uncomplicated acute sinusitis in adults	Amoxicillin	• Amoxicillin-clavulanic acid • Doxycycline • Respiratory FQ (levofloxacin/moxifloxacin) • SMX-TMP • Macrolides • Ceftriaxone
Uncomplicated acute sinusitis in children	Amoxicillin 45-90mg/kg/day	• Amoxicillin-clavulanic acid • Macrolides • SMX-TMP • Cefixime + clindamycin • Doxycycline (>8 years of age)
Risk of antibiotic resistance or failed initial therapy	Amoxicillin-clavulanic acid	Respiratory FQ (Levofloxacin/Moxifloxacin)

How should patients with amoxicillin allergies be managed?

Treatment failure is suspected if symptoms worsen after 72 hours after treatment initiation; reassessment of diagnosis and antibiotic therapy is also required

Duration of therapy is 5-7 days in adults and 10-14 days in children

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Signs and symptoms	Improved and/or Resolved	- Improvement in 2-3 days (fever, maxillary/dental pain, tonsillar exudate, purulent discharge) - Resolution within 1 week
SAFETY ENDPOINTS		
Adverse effects from drug (e.g. diarrhea, upset stomach) and worsening symptoms	Prevent	Duration of therapy

TAKEAWAYS

- Sinusitis is a common upper respiratory infection commonly caused by viruses and less so by bacteria
- The management of most acute sinusitis is supportive care and symptom relief
- Patients who do not improve after 7-10 days or who worsen may benefit from antimicrobial therapy
- Choice of antimicrobial should be directed towards normal respiratory pathogens including *S. pneumoniae*, *M. catarrhalis* and *H. influenzae*
- Duration of therapy for most adult patients can be short (5 – 10 days) but longer in children (10-14 days)

CASE 1

RT is a 34-year-old female who presents with a 3-week history of nasal congestion, facial pain, and green sinus discharge. When her symptoms initially started, she experienced post-nasal drip that progressed to congestion. After 2 weeks of these symptoms, she went to her doctor who prescribed clarithromycin for 7 days, which she reports “didn’t do much.” Today, she comes to the pharmacy with a renewal of her clarithromycin which the doctor has written for an additional week. RT is currently taking levonorgestrol+ethinyl estradiol 1 tablet PO daily (oral contraceptive) and ibuprofen 400 mg PO PRN (for headache – patient reports only taking 1-2 tablets per day). She has a sulfa allergy (rash). She is a mother of two young children.

1. What is the patient likely experiencing?
 - a) Acute bacterial sinusitis
 - b) Chronic bacterial sinusitis
 - c) Acute viral sinusitis
 - d) Chronic viral sinusitis



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2. Which of the following statements regarding the clarithromycin prescription is true?
- a) Clarithromycin is appropriate but requires a longer duration of therapy
 - b) Clarithromycin is not the preferred antibiotic choice for the patient
 - c) The choice of antibiotic and duration are appropriate for the patient
 - d) Another week of clarithromycin should resolve the symptoms



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3. What are acceptable options to treat the patient’s sinusitis?
- a) Amoxicillin and SMX-TMP
 - b) Amoxicillin
 - c) Amoxicillin and ciprofloxacin
 - d) Ciprofloxacin



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4. What is an acceptable recommendation for her symptom management?
- a) Oxymetazoline only
 - b) Acetaminophen/pseudoephedrine only
 - c) Oxymetazoline and diphenhydramine
 - d) Oxymetazoline and acetaminophen/pseudoephedrine



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4. Rosenfeld R. Acute sinusitis in adults. *N Engl J Med* 2016; 375:962-970.
5. Kaplan A. Canadian guidelines for acute bacterial rhinosinusitis: clinical summary. *Can Fam Physician*. 2014;60(3):227-234.

CHANGE LOG

March 2025

- Editorial changes throughout for clarity
- Duration of therapy modified
- No guideline updates available

July 2025

- Slide 16: Amended case questions

October 2025

- No content changes made

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 9: Added note on the PODS criteria, as well as 2 conditions for considering bacterial sinusitis
- Slide 19: Updated references